

讲解

2026 年 7 月 27 日

第一部分

第一小节

第二部分

第二小节

第一章

数学基础

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第二章

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What Are Prime Numbers?

定义

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- ▶ 2 is prime (two divisors: 1 and 2).
- ▶ 3 is prime (two divisors: 1 and 3).
- ▶ 4 is not prime (**three** divisors: 1, 2, and 4).

There Is No Largest Prime Number

The proof uses *reductio ad absurdum*.

定理

There is no largest prime number.

证明.

1. Suppose p were the largest prime number.
2. Consider the number $q = p + 1$.
3. But q is not prime, since q is divisible by p .
4. But $q + 1$ is greater than 1, thus divisible by some prime number not in the first p numbers.



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定理

There is no largest prime number.

证明.

1. Suppose p were the largest prime number.
2. Let q be the product of the first p numbers.
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4. But $q + 1$ is greater than 1, thus divisible by some prime number not in the first p numbers. □

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